ECommerceOrderManagementSystem code Diagram

```
classDiagram
    direction TB

    class Buyer {
        +Products.html
        +ProductsOrder.html
        +ProductsAPI()
        +ProductsShoppingCartAPI()
        +ProductsOrderAPI()
    }

    class Seller {
        +ProductsManagement.html
        +ProductsAPI()
    }

    class Admin {
        +UsersManagement.html
        +UserManagementAPI()
    }

    class UsersAPI {
        +UsersLogin.html
        +login()
    }

    class ProductsAPI {
        +listProducts()
        +getProduct()
        +listProductsBySellerId()
    }

    class ProductsShoppingCartAPI {
        +getShoppingCart()
        +addShoppingCartItem()
        +updateShoppingCartItem()
        +removeShoppingCartItem()
    }

    class ProductsOrderAPI {
        +getOrder()
        +confirmOrder()
        +orderPayment()
    }
```

```
class APIGatewayDefine {
    +permissionCheck()
}

class UserManagementAPI {
    +listUsers()
    +activeUser()
    +inActiveUser()
}

class UserLoginService {
    +login()
}

class PermissionValidationService {
    +getUserPermission()
    +hasPermissionByList()
}

class ProductsService {
    +page_number
    +page_size
    +listProducts()
    +getProductById()
}

class ShoppingCartService {
    +product_id
    +userId
    +getShoppingCart()
    +addShoppingCartItem()
    +updateShoppingCartItem()
    +removeShoppingCartItem()
}

class ProductsOrderPaymentService {
    +orderPayment()
}

class UserManagementService {
    +listUsers()
    +updateUser()
}

class ProductsDB {
    +id
```

```
    +product_name
    +product_price
    +product_size
    +product_color
    +product_sub_name
    +product_description
    +product_img_url
    +product_status
    +listProducts()
    +getProductById()
}

class UsersPermissionRulesDB {
    +id
    +users_id
    +permission_subject_path
    +permission_status
    +hasPermission()
    +getPermissionByUserId()
}

class ProductsDiscountDB {
    +id
    +product_id
    +discount_price
    +discount_status
    +listProductsDiscountByProductsIDs()
}

class ProductsInventoryDB {
    +id
    +product_id
    +inventory_stock
    +inventory_reserved_stock
    +inventory_available_stock
    +inventory_status
    +listProductsInventoryByIDs()
}

class ShoppingCartDB {
    +id
    +product_id
    +users_id
    +products_discount_id
    +products_inventory_id
    +product_name
    +product_price
```

```
    +discount_price
    +product_img_url
    +products_quantity
    +queryProductsShoppingCartByUserId()
    +addProductsShoppingCartByProductId()
    +updateProductsShoppingCartQuantity()
    +delProductsShoppingCartByProductId()
}

class ProductsOrderDB {
    +id
    +users_id
    +order_number
    +order_amount
    +order_discount_price
    +order_product_price
    +order_status
    +payment_status
    +getOrderByIdAndUserIdWithSession()
}

class ProductsOrderItemsDB {
    +id
    +products_order_id
    +users_id
    +products_id
    +products_discount_id
    +products_inventory_id
    +product_name
    +item_quantity
    +item_ordder_number
    +item_product_price
    +item_discount_price
    +item_unit_price
    +getProductsOrderItemsWithSession()
}

class ProductsOrderPaymentDB {
    +id
    +products_order_id
    +users_id
    +payment_number
    +payment_channel
    +payment_type
    +payment_amount
    +payment_order_number
    +payment_status
```

```
    +createProductsOrderPaymentWithSession()
  }

  class UsersDB {
    +id
    +username
    +password
    +display_name
    +user_email
    +user_phone
    +user_status
    +queryUserByUsername()
    +listUsers()
    +queryUserProfileById()
    +updateUser()
  }

  UsersAPI <|-- Buyer : login
  UsersAPI <|-- Seller : login
  UsersAPI <|-- Admin : login

  %% Buyer --> ShoppingCart : manages


  Admin --> APIGatewayDefine : check permission
  %% Admin --> UsersAPI : login
  Buyer --> APIGatewayDefine : check permission
  %% Buyer --> UsersAPI : login
  Seller --> APIGatewayDefine : check permission
  %% Seller --> UsersAPI : login

  Admin --> UserManagementAPI

  Buyer --> ProductsOrderAPI : show orders
  Buyer --> ProductsAPI : show products
  Buyer --> ProductsShoppingCartAPI : shopping cart management

  Seller --> ProductsAPI

  APIGatewayDefine --> PermissionValidationService : check permission
  PermissionValidationService --> UsersPermissionRulesDB : check
permission

  ProductsAPI --> ProductsService : constains
  ProductsShoppingCartAPI --> ShoppingCartService: constains
  ProductsOrderAPI --> ProductsService : constains
  ProductsOrderAPI --> ProductsOrderPaymentService : constains
```

UserManagementAPI --> UserManagementService
UsersAPI --> UserLoginService : login

ShoppingCartService --> ShoppingCartDB : constains
ShoppingCartService --> ProductsDB : constains
ShoppingCartService --> ProductsInventoryDB : constains
ShoppingCartService --> ProductsDiscountDB : constains

ProductsService --> ProductsDB : constains
ProductsService --> ProductsDiscountDB : constains
ProductsService --> ProductsInventoryDB : constains

ProductsOrderPaymentService --> ProductsOrderDB : constains
ProductsOrderPaymentService --> ProductsInventoryDB : constains
ProductsOrderPaymentService --> ProductsOrderItemsDB : constains
ProductsOrderPaymentService --> ProductsOrderPaymentDB : constains

UserManagementService --> UsersDB : constains

UserLoginService --> UsersDB : constains
UserLoginService --> UsersPermissionRulesDB : constains
UserLoginService --> PermissionValidationService : constains